FACIAL EMOTIONS DETECTION

CAPSTONE PROJECT

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PROBLEM DEFINITION

Facial emotions recognition has become crucial for human-machine interactions.

Facial emotion analysis can enhance attendee engagement and help event organizers understand audience sentiment.

55% of communication of sentiment takes place through facial expressions and other facial patterns.

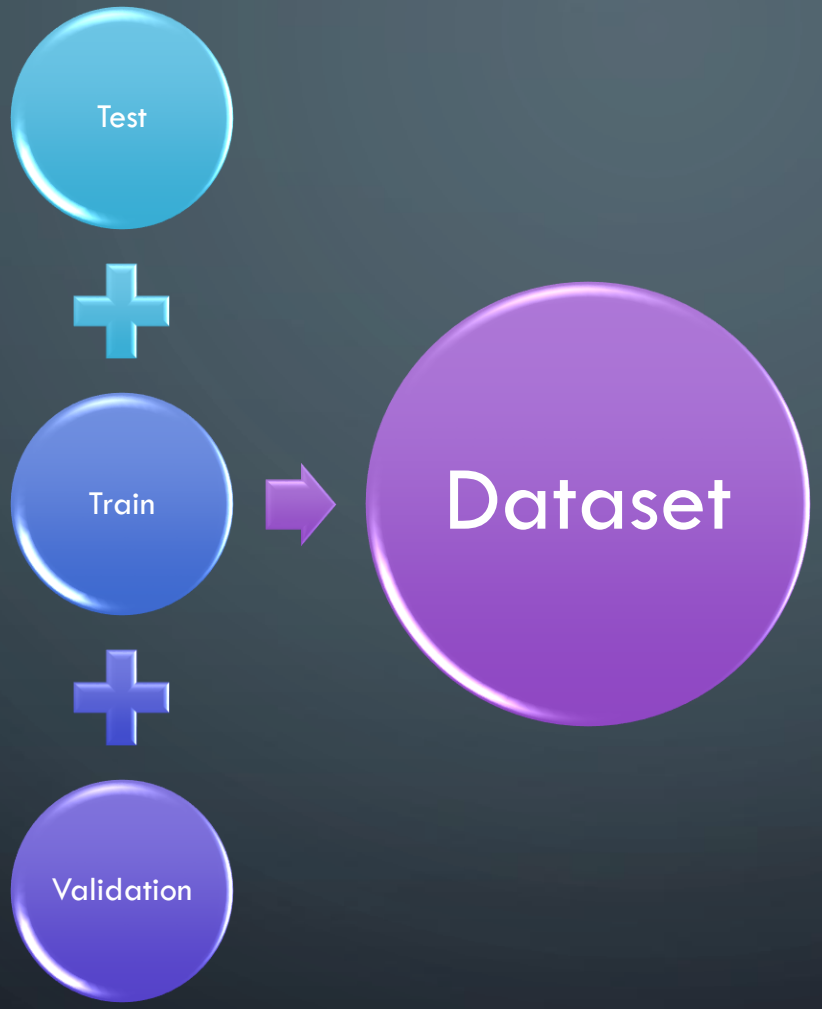
PROBLEM TO SOLVE

To perform sentiment analysis on facial emotions, what are the necessary tools that we can use?

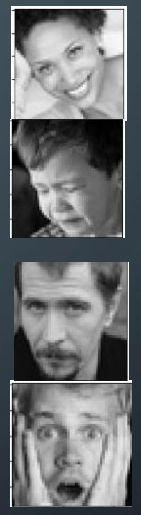
Can we use Deep Learning and AI techniques to create a computer vision model that can accurately detect facial emotions?

Is the model able to classify different facial emotions based on multi-class classification results?

DATA OVERVIEW



HAPPY
SAD
NEUTRAL
SURPRISE



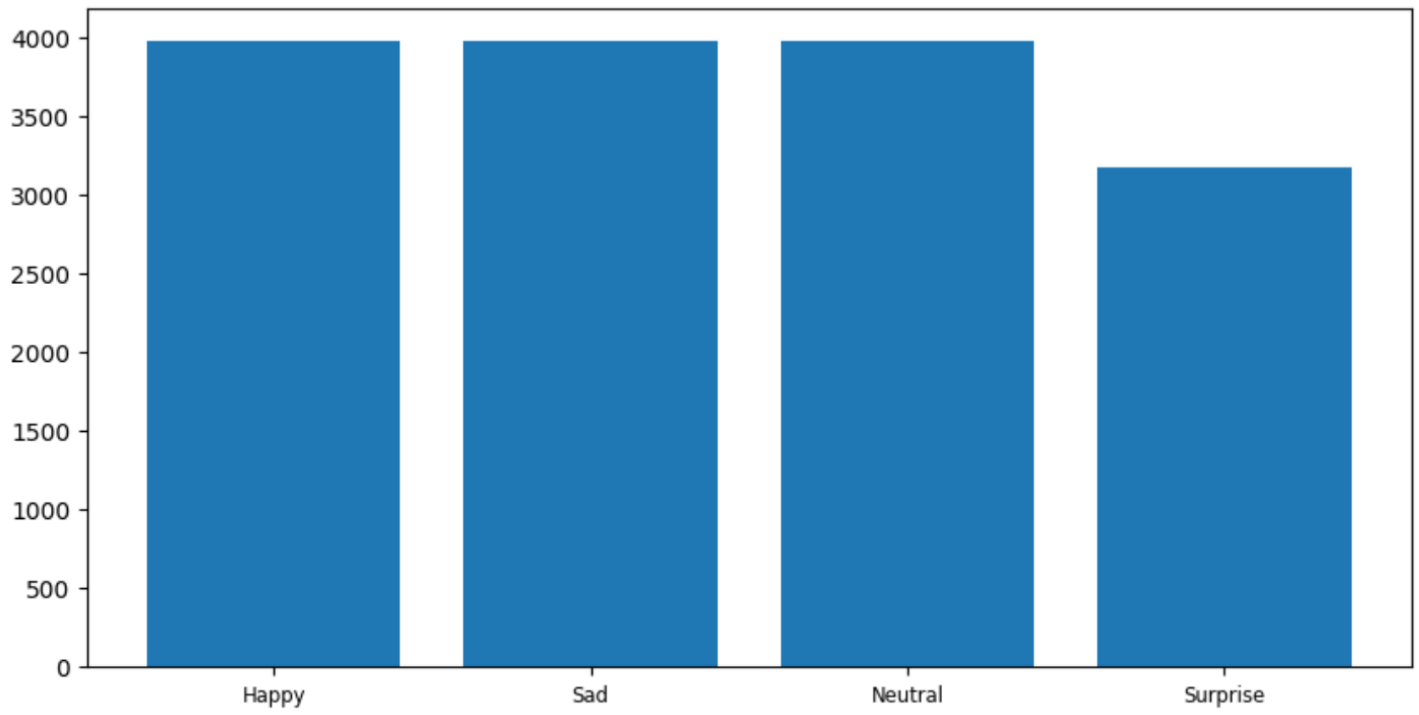
EXPLORATORY DATA ANALYSIS

EMOTION CLASSES DISTRIBUTION

- HAPPY (3976)
- SAD (3982)
- NEUTRAL (3978)
- SURPRISE (3173)

Histogram of 4 emotion classes

- Hapy, Sad & Neutral are evenly distributed.
- Only Surprise is imbalance.

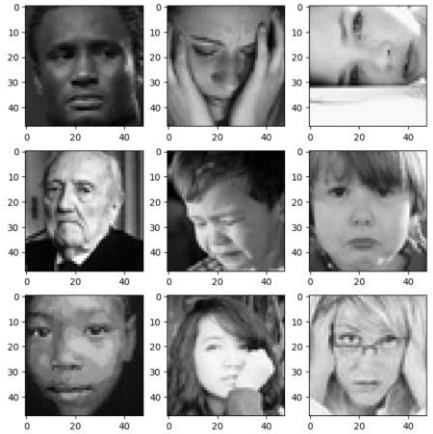


EXPLORATORY DATA ANALYSIS

VISUALIZE DIVERSITY

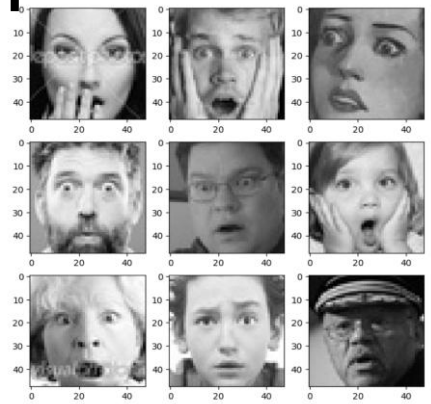
Happy

Sad



Neutral

Surprise



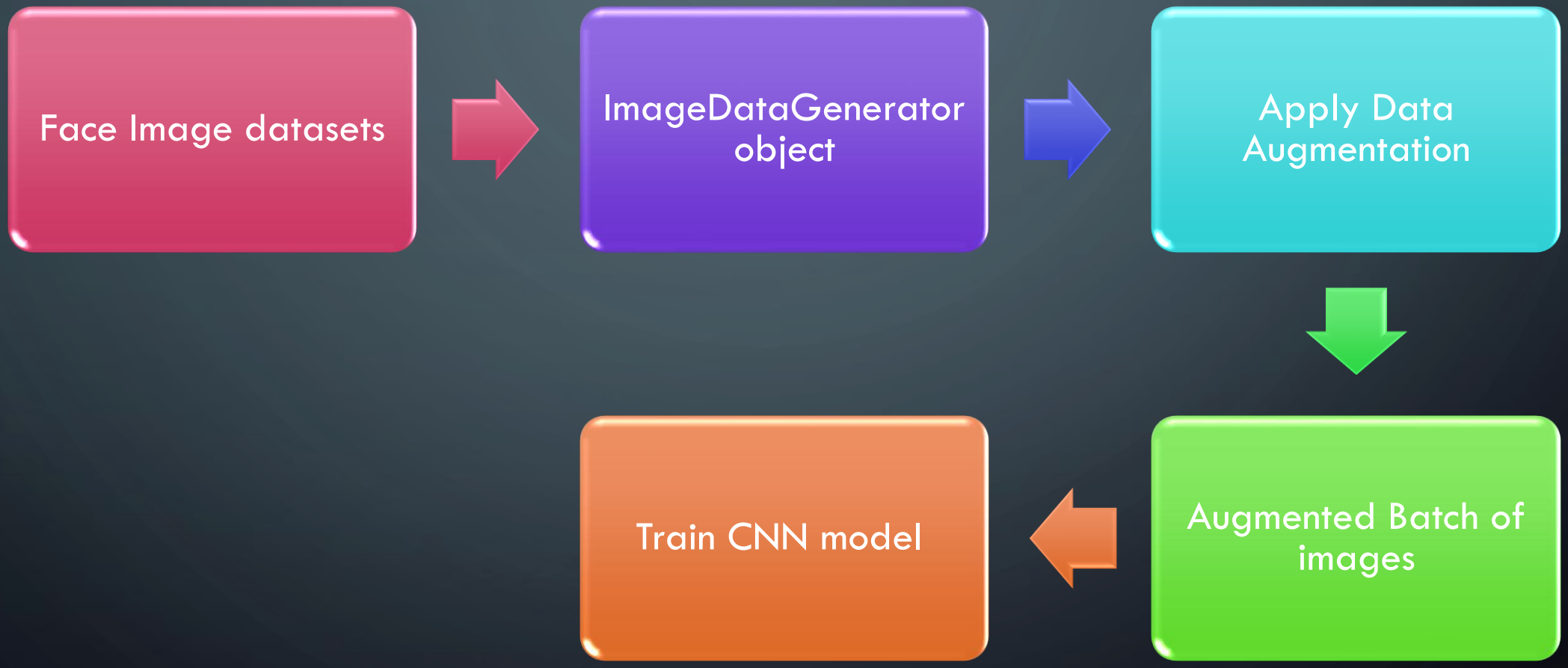
EXPLORATORY DATA ANALYSIS

- **Class Imbalance**
  - Happy, sad and neutral are evenly distributed.
  - Only Surprise is imbalance.
- **Distribution Consistency**
  - Training and testing sets are still consistent.
  - This ensures model's performance could still be evaluated.
- **Image Quality**
  - Varying quality, resolutions, face angles could pose challenges during model training and inference.

DATA PREPROCESSING

- **Data Augmentation**
  - Face Cropping
  - Image Intensity Normalization
  - Enhancement
- **Image Data Generators**
  - Use image data generators for efficient loading and preprocessing of data, reducing memory consumption and enabling the training to handle large datasets.

MODEL COMPARISON

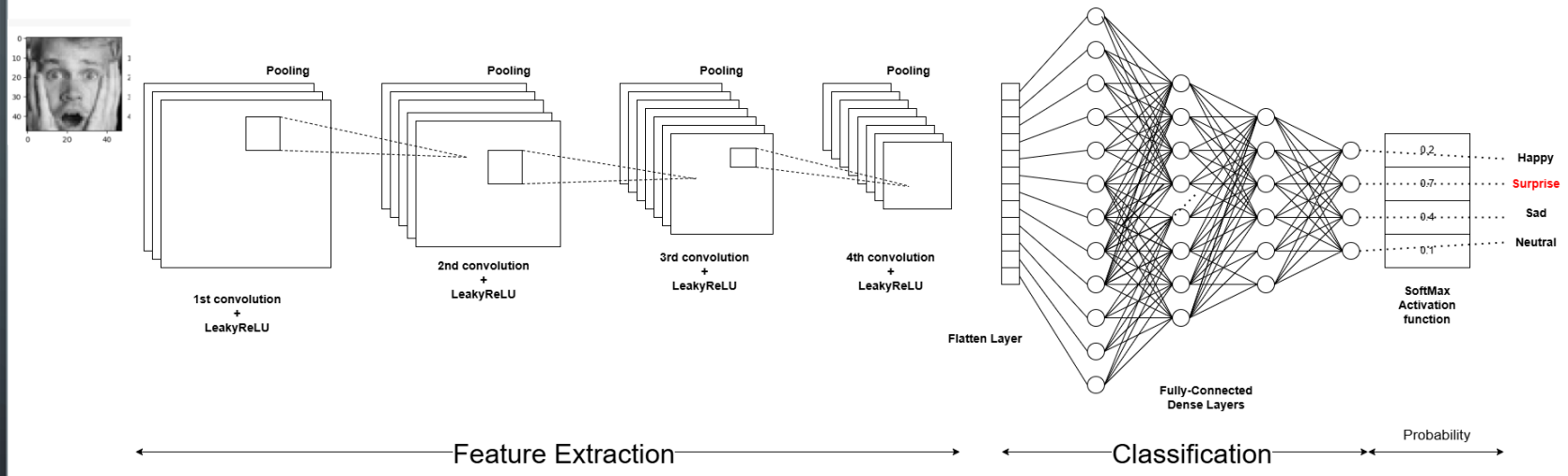


MODEL SOLUTION APPROACH

- Face emotion image datasets
- Exploratory Data Analysis
- Data Pre-processing
- Build, train and tune CNN model
- Model performance metrics and accuracy
- Use the model to classify the emotions

CNN Model Architecture

- Pass an image through **convolutional layers** with **LeakyReLU activations**, which extract complex features.
- These feature maps are then flattened into a single vector, preserving the learned patterns.
- Flows into **fully connected dense layers**, finally, a **SoftMax activation** converts the result into probabilities



MODEL FINE-TUNING

- **Transfer Learning**
  - Apply previously learned model knowledge to a new model.
  - Explore **VGG16, ResNet V2, EfficientNet** models.
- **Callback techniques**
  - Mitigate overfitting and improve model generalization.
  - **ReduceLROnPlateau:** Reduce the learning rate when the training is stagnating.
  - **EarlyStopping:** Stop the training as long as target metric does not improve.

MODEL EVALUATION

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

- **Confusion Matrix**
  - Summarize the performance of the classification model.
  - Compare the predicted classes with actual (true) classes.
- **Metrics and Scoring**
  - Training Accuracy, Validation Accuracy.
  - Precision, recall, f1-score to compare among 4 emotion classes.

MODEL COMPARISON

CNN (66%)

- Base model

VGG16 (49%)

- Simple and consistent
- Heavy computation

ResNet V2

- Allow deep training without vanishing gradients
- Think deep but smartly connected

EfficientNet (25%)

- Balance depth, width and resolution
- Few parameters
- Computationally efficient

FINAL MODEL PROPOSAL

- **ResNet**
  - More efficient and effective about model performance.
  - Deep but Trainable, allow gradients to flow through the network more smoothly.
  - Residual connections help the network reuse features at shallower depths.
  - Robust to overfitting, supporting better model generalization

IMPLEMENTATION RECOMMENDATIONS

- Align ResNet RGB channel input when feeding in new unseen facial images.
- Load my final model using pretrained weights.
- Train and fine-tune model.
- Track accuracy, precision and confusion matrix.
- Use learning curves to monitor model perf

BUSINESS RECOMMENDATIONS

Adjust presentation strategy by analyzing how many happy or sad faces at product launches or live performances.

Adjust content pacing or difficulty level based on signs of facial emotions on virtual classrooms.

Assist healthcare professionals to early detect symtom of depression or anxiety with facial emotion detection over time.

RISKS AND CHALLENGES

- **Risks**
  - People might be recorded or analyzed without consent.
  - Facial emotion detection might tag a person to specific trait that he/she don't agree.
  - Organizations might misuse emotion AI to monitor mental conditions of people.
- **Challenges**
  - Emotion models can mis-intepret facial expressions leading to skewed results.
  - Emotions are complex and nuanced. Smiling faces can also mean angry in heart.
  - If datasets are not diverse enough across demographics, detection will be less accurate and biased towards specific groups.
  - If the model is trained on vivid clear portraits, the model could be easy overfitting when detecting faces in unseen real-world environments.

Q & A